

Exploring Spatial Organization Strategies for Virtual Content in Mixed Reality Environments

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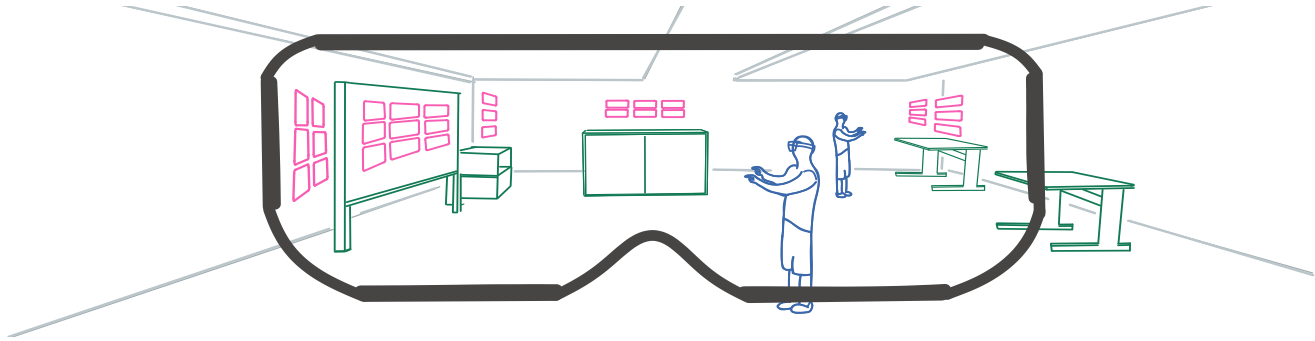


Figure 1: A scenario of organizing mixed-reality content spatially. When arranging virtual content (in pink), users who wear mixed reality headsets are often influenced by their real-world surroundings. Such surroundings consist of 1) physical environments like furniture and architectural elements (in green) and 2) people present (in blue).

ABSTRACT

Our future will likely be reshaped by Mixed Reality (MR) offering boundless display space while preserving the context of real-world surroundings. However, to fully leverage the spatial capabilities of MR technology, a better understanding of how and where to place virtual content like documents is required, particularly considering the situated context. I aim to explore spatial organization strategies for virtual content in MR environments. For that, we conducted empirical studies investigating users' strategies for document layout and placement and examined two real-world factors: physical environments and people present. With this knowledge, we proposed a mixed-reality approach for the in-situ exploration and analysis of human movement data utilizing physical objects in the original space as referents. My next steps include exploring arrangement strategies, designing techniques empowering spatial organization, and extending understandings for multi-user scenarios. My dissertation will enrich the immersive interface repertoire and contribute to the design of future MR systems.

CCS CONCEPTS

• **Human-centered computing** → **Mixed / augmented reality;**
Empirical studies in HCI.

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KEYWORDS

Spatiality, spatial layout, content organization, affordance, Augmented Reality, Mixed Reality

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1 INTRODUCTION & BACKGROUND

Immersive technologies like Mixed Reality (MR) Head-Mounted Displays (HMDs) have the potential to revolutionize our lives. Places like offices will likely be equipped and enhanced by MR HMDs in the future to enrich the current working environment with virtual content. Documents like text, images, spreadsheets, and presentation slides will no longer be confined by rectangle monitors. Instead, holographic counterparts can be placed anywhere in the future office environment, fully leveraging the infinite display space. In fact, various activities have been explored for immersive offices, such as working with general documents [17, 29], exploring brainstorming notes [5, 32], creating design sketches [13, 42], and analyzing data diagrams [15, 19, 20]. Compared to Virtual Reality (VR) that isolates users from reality, MR is more suitable for these everyday office activities, as it maintains the context of real-world surroundings. This supports collective activities with shared awareness [23], allows for easier adoption and integration into the existing workflow [44], and enables the extension and adaption to changeable circumstances [29, 33]. Altogether, these highlight the great potential of MR technology in the workspace.

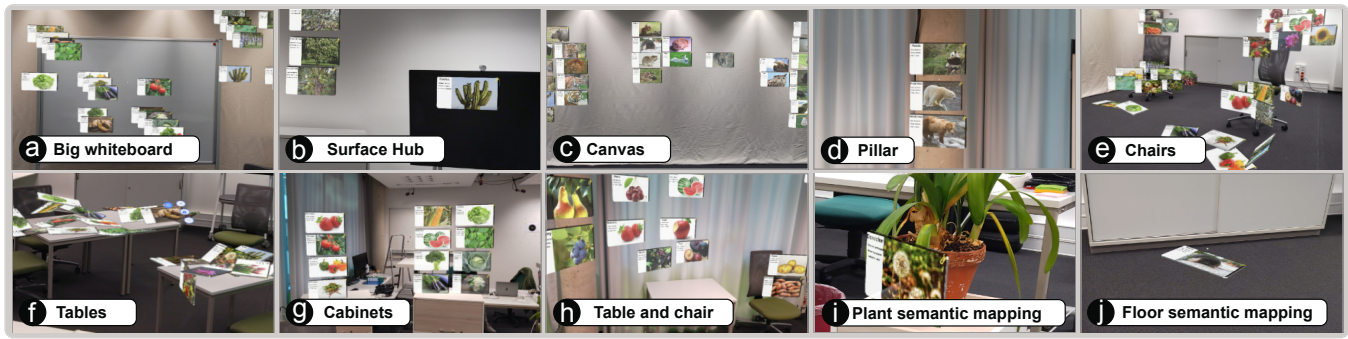


Figure 2: Participants placed MR documents anchoring to furniture and architectural elements for sensemaking in our work [23], including vertical surfaces (a, b, c), horizontal surfaces (f, g, h), the pillar (d), chairs (e), and placement with semantic mapping (i, j).

However, there is a lack of understanding regarding content placement options for future MR interfaces, particularly in the context of real-world surroundings and the spatiality enabled by the output capability. From physical paper [26, 46] to digital files on traditional desktops [1, 3], prior works have proven that users tend to use position and space as means to structure and organize information. For immersive technology, previous works like content organization [15, 18, 39], window layouts [8, 11, 35], and view management [19, 20] in the context of Immersive Analytics (IA) [27] have shown that the data analysis process can benefit from an appropriate design of the content layout. Moreover, several form factors of content layout and placement have been discussed, including the number of visualizations [20], the user task [21, 39], and the dimensionality of the visualizations [15]. Nevertheless, they either only focused on VR environments or lacked considerations of real-world context, which is the inherent nature of MR. Particularly, the effect of physical environments on MR content layout and placement is still unclear. Affordances of objects like furniture, considering its geometry and semantics, have not been sufficiently discussed. Moreover, although activities like brainstorming, sensemaking, and presentation are intrinsically multi-user, adapting layout and placement to the involvement of people present has seldom been examined.

Related works have suggested integrating and contextualizing holographic content (e.g., data visualizations [16, 47]) into reality by placing it near or embedded in physical objects. Furthermore, locations like physical surfaces can be the anchors for content placement [31], such as aligning to ceiling and floor [37] or other 2D displays [14, 22, 34]. Moreover, generic heuristics have been considered to support placement decisions in the real world, including visual salience, spatial consistency [10], and real-world backgrounds [36]. Aside from this, the semantic association between virtual content and physical environments has also been considered for photo presentation [2] and general interface layout adaptation [4] in MR. For a working scenario, Spatial Analytic Interface [9] advocated that users should work in situ where information is semantically associated. Recent works further explored this topic concerning how to present and arrange content (like charts) in relation to its physical referent [16, 40] as well as the

pros and cons of different coordinating layouts [38, 45]. In addition to physical objects, related work (e.g., mirrored layout [13] or subjective views [41]) has also considered people present in the environment for view arrangements. Lastly, several techniques have been demonstrated to facilitate the arrangement process, such as automatic alignment [4, 31], semi-automated arrangement [30, 43], and optimal area detection [28]. These works highlight the necessity to consider the real-world surroundings and situated contexts for the spatial organization of MR content. Nevertheless, knowledge of presenting, placing, and arranging information in MR and corresponding design guidelines is still lacking [7].

2 OVERVIEW OF RESEARCH QUESTIONS

I aim to contribute a better understanding of content organization options and factors in designing future MR systems. To clarify, I describe the spatial organization of mixed-reality content as an activity using certain arrangement methods to achieve the layout and placement of multiple pieces of content in a given real-world surrounding (see Fig. 1).

With this, my dissertation explores the following research questions:

RQ1: How would users place and organize virtual content considering real-world surroundings? I first investigated how users organize virtual documents in a given real-world context. My aim was to comprehend how the presence of people and physical surroundings influence the placement and arrangement of content. This sets up an overarching foundation for my dissertation.

RQ2: How can designers utilize physical environments for organizing virtual content? Inspired by the results addressing RQ1, I explored the utilization of physical environments for organizing virtual content, particularly considering affordances suggested by geometry and semantic characteristics of environments.

RQ3: How would users arrange virtual content for the organization tasks? With varying purposes, organizational activities like grouping and spreading can be dynamic and require user involvement in adjusting content placement and layout. We want to understand user practice and support this workflow with interaction techniques.

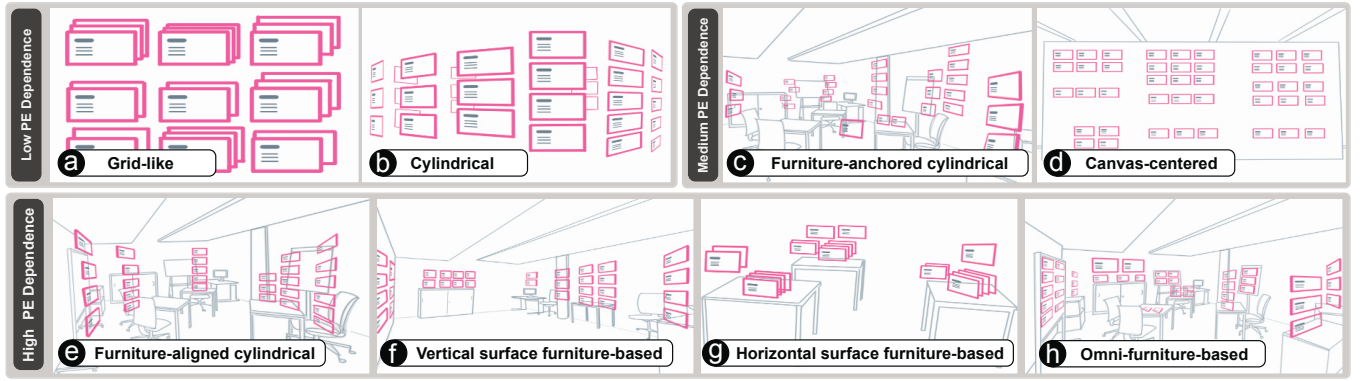


Figure 3: Participants achieved various layouts to organize MR documents spatially in our work [23]. We categorized these strategies based on the degree of dependence on the physical environment (PE) from low (a-b), medium (c-d), to high (e-h).

RQ4: How can virtual content be placed and organized considering the people present? Real-world surroundings consist of not only relatively static entities like physical objects, but also the presence of people. In particular, depending on the involvement in the ongoing activity, these people can be collaborators with similar goals, users with different roles, or even bystanders. Thus, we aim to investigate content organization considering the dynamicity of multiple users and their social context.

3 RESEARCH TO DATE

So far, several projects have been conducted, and results have been published in CHI '22 and '23. I applied a number of methodologies for these projects, including literature review, qualitative user studies like lab-controlled experiments and expert evaluations, visualization and interaction design, and prototype development. Specifically, we investigated virtual document placement and layout strategies and examined the real-world factors (RQ1) [6, 23, 24]. Furthermore, we presented PEARL [25], a mixed-reality approach for analyzing human movement in situ, exemplifying how designers and researchers can utilize the physical environments with MR content (RQ2).

Understanding Real-world Surroundings (RQ1). MR could likely benefit office activities, especially for collaborative brainstorming and sensemaking. However, it is unclear how physical environments and co-located collaboration influence the spatial organization of virtual content for sensemaking. For this, an initial user study (N=8) [24] and a full-extent user study (N=28) [23] were conducted. We investigated the effect of office environments (fully-furnished vs. partly-furnished) and work styles (individual vs. collaborative) during a document classification task using Augmented Reality (AR) with regard to content placement, layout strategies, and sensemaking workflows. Results showed that furniture like tables and whiteboards were required to assist content organization for sensemaking and collaboration regardless of room settings (see Fig. 2). Moreover, furniture usage was prioritized in the collaborative context despite personal layout preferences. We identified different layout strategies, from vertical and horizontal surface furniture-based layouts that highly depend on physical environments, to grid and cylindrical layouts that are little related

to physical environments (see Fig. 3). Lastly, we proposed design implications to guide the design of MR applications.

Moreover, we further investigated how situated environments influence user perception of virtual content layout through another user study (N=8) [6]. Based on this, we theorized the relation between virtual content and its real-world context by introducing a two-dimensional design space (spatial and semantic coupling) with associated parameters (e.g., position and rotation).

Our works were the first attempts to reveal the organization strategies of virtual content with a focus on real-world surroundings. In particular, for organizing virtual content in MR, physical environments and people present are crucial aspects to be considered. Based on this foundation, I further develop my dissertation with the following research.

Utilizing Physical Environments (RQ2). To demonstrate the application of gained knowledge in RQ1, we introduced PEARL [25], a mixed-reality approach for the analysis of human movement data in situ. In indoor environments, human movement is often affected by physical objects populating a space and their affordances. A physical object might be an obstacle that has to be avoided to continue the movement or an object of interest that is intentionally approached to interact with it. Thus, physical environments restrict, shape, or elicit human behavior and motion. Hence, analyzing such motion can benefit from the direct inclusion of the environment in the analytical process.

PEARL contains methods for exploring movement data in relation to surrounding regions of interest, such as objects, furniture, and architectural elements. We introduced concepts for selecting and filtering data through direct interaction with the environment. Besides, a suite of visualizations (see Fig. 4) was designed to be associated with physical referents for revealing aggregated and emergent spatial and temporal relations. To illustrate its potential, an AR-based prototype was developed and reviewed via expert evaluations and scenario walkthroughs (N=4) in a simulated exhibition. We systematically analyzed the results and reported the first insights of our approach. PEARL lays a foundation for utilizing physical environments in the in-situ analysis of movement data.

This work is a demonstration of the design guidelines derived from RQ1. In particular, it illustrates how to position virtual content

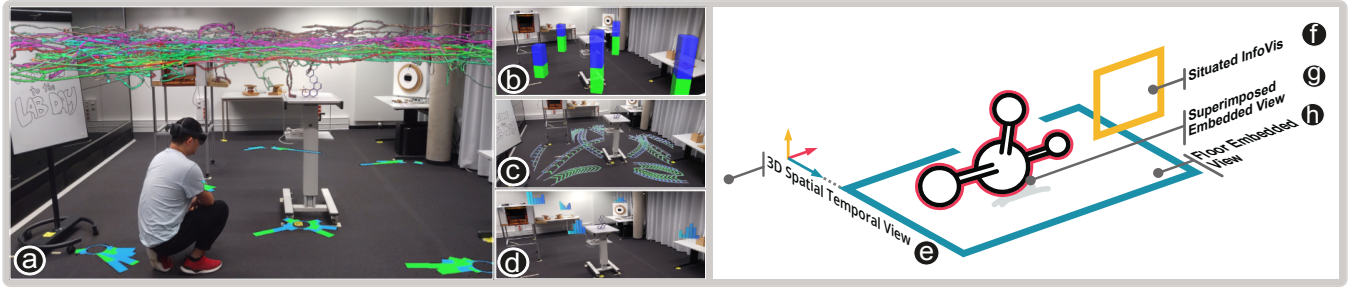


Figure 4: Our PEARL prototype with various MR visualizations anchoring to physical objects (a-d) for analyzing human movement and a placement scheme organizing four types of MR visualizations (e-h). 3D trajectories showing human movement are placed in the open space in (a) and (e); 3D bars superimpose on the associated physical objects in (b) and (g); movement flows are embedded on the floor in (c) and (h); 2D information visualization panels are placed side-by-side in (d) and (f).

in physical environments and the value of utilizing such environments for designing MR systems, thus answering RQ2.

4 NEXT STEPS

Next, I plan to explore arrangement strategies and interaction techniques for organizing virtual content (RQ3). Moreover, I aim to extend the understanding of real-world surroundings by focusing on the people present (RQ4).

Exploring Spatial Arrangement Strategies (RQ3). Participants in our studies [23, 24] appreciated the potential and benefits of being able to arrange virtual documents spatially, even with simple direct interaction and ray-casting interaction. They often rearranged document layout and placement, resulting in a dynamic and iterative process that requires constant actions such as clustering, translating, and spreading documents. Therefore, we want to investigate further these arrangement practices and strategies concerning general document organization tasks. In line with human-centered design, we aim to design interaction techniques to facilitate organizational workflows. Ultimately, we plan to establish a design framework that leverages natural human interaction to guide the design of such systems.

For this, the following research questions are worth highlighting. Specifically, **(1) strategies for creating and managing spatial layouts:** How would users cluster documents into a particular layout? How would this layout be inspected, translated, and modified? **(2) the interplay between arrangement strategies and content placement:** While the spatial layout can be constructed by arrangement actions, would document layout and placement also inform and result in arrangement strategies? **(3) general interaction style and modality:** What input modality would be preferred for arranging content? Particularly, how can physical environments and objects be utilized for arrangement actions? To understand these questions, we have conducted a participatory design study, and the results are being analyzed.

Investigating Multi-user Context (RQ4). We found that collaborating participants tend to use furniture for organizing content, some even overriding their personal preference [23]. Thus, social

context seems influential in content organization and could potentially be weighted more than other factors under certain circumstances. This highlights the importance of RQ4 to explore factors considering the variety of social contexts and their impacts. Moreover, considering the presence of people in the environment and empowering multi-user scenarios would align with the core and unique value of Mixed Reality. In particular, beyond expanding display pixels to show more content, this technology could advance both interaction with computing devices and interaction among human beings.

Despite this ambition, the complexity of social context challenges the design of future MR systems, where I want to highlight two particular research avenues. First, while our previous work [23] has contributed to a collaboration scenario, follow-up investigations are needed. For instance, existing research [13] often only considered a limited number of collaborators, while real-world situations could be more complex. Besides, collaboration activities can also be dynamic with changes like collaborators joining or exiting. In this case, content placement and layout need to adapt to such an alteration, e.g., adjust the content orientation to maintain collective visibility. In addition, the placement and layout of content will likely be contextualized by the relationships of collaborators, in particular, considering the formation of distance among collaborators, or proxemics [12].

Aside from a typical collaboration scenario where collaborators share the same or similar goals, involved people could have different roles, such as in a presentation scenario. MR technology has great potential for presentation. In this case, users often engage with different purposes, resulting in distributed roles, i.e., presenters and audiences. Therefore, immersive content requires to be organized to assist not only in presenting content but also in the comprehension of information. Content layout and placement could enhance the engagement of audiences but could also undermine the experience as distractions. Understanding this spatial relation and the interplay among immersive content, the presenter, and the audience would be critical for the design of such MR systems. I am planning to approach these questions with methods like empirical studies, visualization and interaction design, and prototype development.

5 DISSERTATION STATUS AND LONG-TERM GOALS

I am currently a fourth-year Ph.D. student in the Interactive Media Lab at the Faculty of Computer Science, Technische Universität Dresden, supervised by Professor Raimund Dachsel. I have not attended previous doctoral symposiums. For this research topic, I have published four papers, including two CHI full papers [23, 25] and two Late-Breaking Works (LBWs) [6, 24], and I aim to complete the proposed research projects and the dissertation before Spring 2025. After graduation, I plan to seek research positions in academia to investigate mixed-reality interface design further and advocate for future MR-empowered workspaces.

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REFERENCES

- [1] Christopher Andrews, Alex Endert, and Chris North. 2010. Space to Think: Large High-Resolution Displays for Sensemaking. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Atlanta, Georgia, USA) (CHI '10). Association for Computing Machinery, New York, NY, USA, 55–64. <https://doi.org/10.1145/1753326.1753336>
- [2] Han Joo Chae, Youli Chang, Minji Kim, Gwanmo Park, and Jinwook Seo. 2020. ARphy: Managing Photo Collections Using Physical Objects in AR. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI EA '20). Association for Computing Machinery, New York, NY, USA, 1–7. <https://doi.org/10.1145/3334480.3382885>
- [3] Chia-Ming Chang, Chia-Hsien Lee, and Takeo Igarashi. 2021. Spatial Labeling: Leveraging Spatial Layout for Improving Label Quality in Non-Expert Image Annotation. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 306, 12 pages. <https://doi.org/10.1145/3411764.3445165>
- [4] Yifei Cheng, Yukang Yan, Xin Yi, Yuanchun Shi, and David Lindlbauer. 2021. SemanticAdapt: Optimization-Based Adaptation of Mixed Reality Layouts Leveraging Virtual-Physical Semantic Connections. In *The 34th Annual ACM Symposium on User Interface Software and Technology* (Virtual Event, USA) (UIST '21). Association for Computing Machinery, New York, NY, USA, 282–297. <https://doi.org/10.1145/3472749.3474750>
- [5] Kylie Davidson, Lee Lisle, Kirsten Whitley, Doug A. Bowman, and Chris North. 2022. Exploring the Evolution of Sensemaking Strategies in Immersive Space to Think. *IEEE Transactions on Visualization and Computer Graphics* (2022), 1–15. <https://doi.org/10.1109/TVCG.2022.3207357>
- [6] Mats Ole Ellenberg, Marc Satkowski, Weizhou Luo, and Raimund Dachsel. 2023. Spatiality and Semantics - Towards Understanding Content Placement in Mixed Reality. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI EA '23). Association for Computing Machinery, New York, NY, USA, Article 254, 8 pages. <https://doi.org/10.1145/3544549.3585853>
- [7] Barrett Ens, Benjamin Bach, Maxime Cordeil, Ulrich Engelke, Marcos Serrano, Wesley Willett, Arnaud Prouzeau, Christoph Anthes, Wolfgang Büschel, Cody Dunne, Tim Dwyer, Jens Grubert, Jason H. Haga, Nurit Kirshenbaum, Dylan Kobayashi, Tica Lin, Monsurat Olaosebikan, Fabian Pointecker, David Saffo, Nazmus Saquib, Dieter Schmalstieg, Danielle Albers Szafir, Matt Whitlock, and Yalong Yang. 2021. Grand Challenges in Immersive Analytics. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 459, 17 pages. <https://doi.org/10.1145/3411764.3446866>
- [8] Barrett Ens, Juan David Hincapié-Ramos, and Pourang Irani. 2014. Ethereal planes: a design framework for 2D information space in 3D mixed reality environments. In *Proceedings of the 2nd ACM symposium on Spatial user interaction*. ACM, Honolulu Hawaii USA, 2–12. <https://doi.org/10.1145/2659766.2659769>
- [9] Barrett Ens and Pourang Irani. 2017. Spatial Analytic Interfaces: Spatial User Interfaces for In Situ Visual Analytics. *IEEE Computer Graphics and Applications* 37, 2 (2017), 66–79. <https://doi.org/10.1109/MCG.2016.38>
- [10] Barrett Ens, Eyal Ofek, Neil Bruce, and Pourang Irani. 2015. Spatial Constancy of Surface-Embedded Layouts across Multiple Environments. In *Proceedings of the 3rd ACM Symposium on Spatial User Interaction* (Los Angeles, California, USA) (SUI '15). Association for Computing Machinery, New York, NY, USA, 65–68. <https://doi.org/10.1145/2788940.2788954>
- [11] Barrett M. Ens, Rory Finnegan, and Pourang P. Irani. 2014. The Personal Cockpit: A Spatial Interface for Effective Task Switching on Head-Worn Displays. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Toronto, Ontario, Canada) (CHI '14). Association for Computing Machinery, New York, NY, USA, 3171–3180. <https://doi.org/10.1145/2556288.2557058>
- [12] Edward T Hall, Ray L Birdwhistell, Bernhard Bock, Paul Bohannon, A Richard Diebold Jr, Marshall Durbin, Munro S Edmonson, JL Fischer, Dell Hymes, Solon T Kimball, et al. 1968. Proxemics [and comments and replies]. *Current anthropology* 9, 2/3 (1968), 83–108.
- [13] Zhenyi He, Ruofei Du, and Ken Perlin. 2020. CollaboVR: A Reconfigurable Framework for Creative Collaboration in Virtual Reality. In *2020 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. 542–554. <https://doi.org/10.1109/ISMAR50242.2020.00082>
- [14] Ricardo Langner, Marc Satkowski, Wolfgang Büschel, and Raimund Dachsel. 2021. MARVIS: Combining Mobile Devices and Augmented Reality for Visual Data Analysis. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 468, 17 pages. <https://doi.org/10.1145/3411764.3445593>
- [15] Benjamin Lee, Xiaoyun Hu, Maxime Cordeil, Arnaud Prouzeau, Bernhard Jenny, and Tim Dwyer. 2021. Shared Surfaces and Spaces: Collaborative Data Visualisation in a Co-located Immersive Environment. *IEEE Transactions on Visualization and Computer Graphics* 27, 2 (2021), 1171–1181. <https://doi.org/10.1109/TVCG.2020.3030450>
- [16] Benjamin Lee, Michael Sedlmair, and Dieter Schmalstieg. 2023. Design Patterns for Situated Visualization in Augmented Reality. *IEEE Transactions on Visualization and Computer Graphics* (2023), 1–12. <https://doi.org/10.1109/TVCG.2023.3327398>
- [17] Zhen Li, Michelle Annett, Ken Hinckley, Karan Singh, and Daniel Wigdor. 2019. HoloDoc: Enabling Mixed Reality Workspaces That Harness Physical and Digital Content. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland Uk) (CHI '19). Association for Computing Machinery, New York, NY, USA, 1–14. <https://doi.org/10.1145/3290605.3300917>
- [18] Lee Lisle, Xiaoyu Chen, J. K. Edward Gitre, Chris North, and Doug A. Bowman. 2020. Evaluating the Benefits of the Immersive Space to Think. In *2020 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*. IEEE. <https://doi.org/10.1109/vrw50115.2020.00073>
- [19] Jiazhou Liu, Barrett Ens, Arnaud Prouzeau, Jim Smiley, Isobel Kara Nixon, Sarah Goodwin, and Tim Dwyer. 2023. DataDancing: An Exploration of the Design Space For Visualisation View Management for 3D Surfaces and Spaces. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 379, 17 pages. <https://doi.org/10.1145/3544548.3580827>
- [20] Jiazhou Liu, Arnaud Prouzeau, Barrett Ens, and Tim Dwyer. 2020. Design and Evaluation of Interactive Small Multiples Data Visualisation in Immersive Spaces. In *2020 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*. 588–597. <https://doi.org/10.1109/VR46266.2020.00081>
- [21] Jiazhou Liu, Arnaud Prouzeau, Barrett Ens, and Tim Dwyer. 2022. Effects of Display Layout on Spatial Memory for Immersive Environments. *Proc. ACM Hum.-Comput. Interact.* 6, ISS, Article 576, 21 pages. <https://doi.org/10.1145/3567729>
- [22] Feiyu Lu and Doug A. Bowman. 2021. Evaluating the Potential of Glanceable AR Interfaces for Authentic Everyday Uses. In *2021 IEEE Virtual Reality and 3D User Interfaces (VR)*. 768–777. <https://doi.org/10.1109/VR50410.2021.00104>
- [23] Weizhou Luo, Anke Lehmann, Hjalmar Widengren, and Raimund Dachsel. 2022. Where Should We Put It? Layout and Placement Strategies of Documents in Augmented Reality for Collaborative Sensemaking. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI '22). Association for Computing Machinery, New York, NY, USA, Article 627, 16 pages. <https://doi.org/10.1145/3491102.3501946>
- [24] Weizhou Luo, Anke Lehmann, Yushan Yang, and Raimund Dachsel. 2021. Investigating Document Layout and Placement Strategies for Collaborative Sensemaking in Augmented Reality. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI EA '21). Association for Computing Machinery, New York, NY, USA, Article 456, 7 pages. <https://doi.org/10.1145/3411763.3451588>
- [25] Weizhou Luo, Zhongyuan Yu, Rufat Rzaev, Marc Satkowski, Stefan Gumhold, Matthew McGinity, and Raimund Dachsel. 2023. Pearl: Physical Environment Based Augmented Reality Lenses for In-Situ Human Movement Analysis. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 381, 15 pages. <https://doi.org/10.1145/3544548.3580715>
- [26] Thomas W. Malone. 1983. How Do People Organize Their Desks? Implications for the Design of Office Information Systems. *ACM Trans. Inf. Syst.* 1, 1 (jan 1983),

- 99–112. <https://doi.org/10.1145/357423.357430>
- [27] Kim Marriott, Falk Schreiber, Tim Dwyer, Karsten Klein, Nathalie Henry Riche, Takayuki Itoh, Wolfgang Stuerzlinger, and Bruce H Thomas. 2018. *Immersive analytics*. Vol. 11190. Springer.
- [28] Fabrice Matulic, Wolfgang Büschel, Michael Ying Yang, Stephan Ihrke, Anmol Ramraika, Carsten Rother, and Raimund Dachselt. 2016. Smart Ubiquitous Projection: Discovering Surfaces for the Projection of Adaptive Content. In *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems* (San Jose, California, USA) (CHI EA '16). Association for Computing Machinery, New York, NY, USA, 2592–2600. <https://doi.org/10.1145/2851581.2892545>
- [29] Daniel Medeiros, Mark McGill, Alexander Ng, Robert McDermid, Nadia Pantidi, Julie Williamson, and Stephen Brewster. 2022. From Shielding to Avoidance: Passenger Augmented Reality and the Layout of Virtual Displays for Productivity in Shared Transit. *IEEE Transactions on Visualization and Computer Graphics* 28, 11 (2022), 3640–3650. <https://doi.org/10.1109/TVCG.2022.3203002>
- [30] Aziz Niyazov, Barrett Ens, Kadek Ananta Satriadi, Nicolas Mellado, Loic Barthe, Tim Dwyer, and Marcos Serrano. 2023. User-Driven Constraints for Layout Optimisation in Augmented Reality. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (CHI '23). Association for Computing Machinery, New York, NY, USA, 1–16. <https://doi.org/10.1145/3544548.3580873>
- [31] Benjamin Nuernberger, Eyal Ofek, Hrvoje Benko, and Andrew D. Wilson. 2016. SnapToReality: Aligning Augmented Reality to the Real World. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems* (San Jose, California, USA) (CHI '16). Association for Computing Machinery, New York, NY, USA, 1233–1244. <https://doi.org/10.1145/2858036.2858250>
- [32] Monsurat Olaosebikan, Claudia Aranda Barrios, Lenore Cowen, and Orit Shaer. 2022. Embodied Notes: A Cognitive Support Tool For Remote Scientific Collaboration in VR. In *Extended Abstracts of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI EA '22). Association for Computing Machinery, New York, NY, USA, Article 279, 6 pages. <https://doi.org/10.1145/3491101.3519664>
- [33] Leonardo Pavanatto, Chris North, Doug A. Bowman, Carmen Badea, and Richard Stokley. 2021. Do we still need physical monitors? An evaluation of the usability of AR virtual monitors for productivity work. In *2021 IEEE Virtual Reality and 3D User Interfaces (VR)*. 759–767. <https://doi.org/10.1109/VR50410.2021.00103>
- [34] Patrick Reipschläger and Raimund Dachselt. 2019. DesignAR: Immersive 3D-Modeling Combining Augmented Reality with Interactive Displays. In *Proceedings of the 2019 ACM International Conference on Interactive Surfaces and Spaces* (Daejeon, Republic of Korea) (ISS '19). Association for Computing Machinery, New York, NY, USA, 29–41. <https://doi.org/10.1145/3343055.3359718>
- [35] Jie Ren, Yueting Weng, Chengchi Zhou, Chun Yu, and Yuanchun Shi. 2020. Understanding Window Management Interactions in AR Headset + Smartphone Interface. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems*. ACM. <https://doi.org/10.1145/3334480.3382812>
- [36] Marc Satkowski and Raimund Dachselt. 2021. Investigating the Impact of Real-World Environments on the Perception of 2D Visualizations in Augmented Reality. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 522, 15 pages. <https://doi.org/10.1145/3411764.3445330>
- [37] Marc Satkowski, Rufat Rzaev, Eva Goebel, and Raimund Dachselt. 2022. ABOVE & BELOW: Investigating Ceiling and Floor for Augmented Reality Content Placement. In *2022 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. IEEE, 518–527. <https://doi.org/10.1109/ISMAR55827.2022.00068>
- [38] Kadek Ananta Satriadi, Andrew Cunningham, Bruce H. Thomas, Adam Drogemuller, Antoine Odi, Niki Patel, Cathlyn Aston, and Ross T. Smith. 2022. Augmented Scale Models: Presenting Multivariate Data Around Physical Scale Models in Augmented Reality. In *2022 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. 54–63. <https://doi.org/10.1109/ISMAR55827.2022.00019>
- [39] Kadek Ananta Satriadi, Barrett Ens, Maxime Cordeil, Tobias Czauderna, and Bernhard Jenny. 2020. Maps Around Me: 3D Multiview Layouts in Immersive Spaces. *Proc. ACM Hum.-Comput. Interact.* 4, ISS, Article 201 (nov 2020), 20 pages. <https://doi.org/10.1145/3427329>
- [40] Kadek Ananta Satriadi, Jim Smiley, Barrett Ens, Maxime Cordeil, Tobias Czauderna, Benjamin Lee, Ying Yang, Tim Dwyer, and Bernhard Jenny. 2022. Tangible Globes for Data Visualisation in Augmented Reality. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI '22). Association for Computing Machinery, New York, NY, USA, Article 505, 16 pages. <https://doi.org/10.1145/3491102.3517715>
- [41] Mickael Sereno and Tobias Isenberg. 2020. Subjective Views in Co-Located Augmented Reality-Initial Design. In *Posters of IEEE Visualization*.
- [42] Ryo Suzuki, Rubaiat Habib Kazi, Li-yi Wei, Stephen DiVerdi, Wilmot Li, and Daniel Leithinger. 2020. RealitySketch: Embedding Responsive Graphics and Visualizations in AR through Dynamic Sketching. In *Proceedings of the 33rd Annual ACM Symposium on User Interface Software and Technology*. ACM, Virtual Event USA, 166–181. <https://doi.org/10.1145/3379337.3415892>
- [43] Ibrahim A. Tahmid, Lee Lisle, Kylie Davidson, Chris North, and Doug A. Bowman. 2022. Evaluating the Benefits of Explicit and Semi-Automated Clusters for Immersive Sensemaking. In *2022 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. 479–488. <https://doi.org/10.1109/ISMAR55827.2022.00064> ISSN: 1554-7868.
- [44] Xiyao Wang, Lonni Besançon, David Rousseau, Mickael Sereno, Mehdi Ammi, and Tobias Isenberg. 2020. Towards an Understanding of Augmented Reality Extensions for Existing 3D Data Analysis Tools. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '20). Association for Computing Machinery, New York, NY, USA, 1–13. <https://doi.org/10.1145/3313831.3376657>
- [45] Zhen Wen, Wei Zeng, Luoxuan Weng, Yihan Liu, Mingliang Xu, and Wei Chen. 2023. Effects of View Layout on Situated Analytics for Multiple-View Representations in Immersive Visualization. *IEEE Transactions on Visualization and Computer Graphics* 29, 1 (2023), 440–450. <https://doi.org/10.1109/TVCG.2022.3209475>
- [46] John Wenskovitch and Chris North. 2021. An Examination of Grouping and Spatial Organization Tasks for High-Dimensional Data Exploration. *IEEE Transactions on Visualization and Computer Graphics* 27, 2 (2021), 1742–1752. <https://doi.org/10.1109/TVCG.2020.3028890>
- [47] Wesley Willett, Yvonne Jansen, and Pierre Dragicevic. 2017. Embedded Data Representations. *IEEE Transactions on Visualization and Computer Graphics* 23, 1 (2017), 461–470. <https://doi.org/10.1109/TVCG.2016.2598608>